

☎ CONTACT

Bangalore, India
+91 7069321279
chiragjivani1812@gmail.com
linkedin.com/in/chirag-jivani

★ AT A GLANCE

7+ years in software & automation
3+ years on railway TCMS
10 train & fieldbus protocols
4 countries: IN · FR · RO · CO

⚙ TECHNICAL SKILLS

Programming
C, C++, Python, Shell scripting (Bash), IEC 61131-3 (ST/SCL, FBD, Ladder), VBScript, PHP, HTML/JS

Operating Systems
Linux – Yocto, Debian

Debug & Analysis
GDB (core/crash dumps), LTTng trace analysis, Wireshark, Docklight, log analysis

Railway / AGATE
Alchemist, Diagnostics Editor, OCoCheTo, Dataplug Manager, ControlBuild, CODESYS

Automation & SCADA
TIA Portal V16, SIMATIC Manager V5.6, WinCC Explorer V7.5, SCOUT, WPLSoft, Node-RED

Version Control
GitLab (commits, merge requests, code review), IBM ClearCase / ClearQuest, Redmine

Other
Arduino, MATLAB, MS SQL Server

⇄ PROTOCOLS

TRDP · CIP/EtherNet-IP · MVB
CAN · HPTS · Modbus TCP/RTU
OPC UA · RS-232/485 · I2C · SPI

CHIRAG JIVANI

> Senior Software Engineer_

Railway TCMS · C Middleware · Linux (Yocto / Debian) · Real-Time Diagnostics · Industrial Automation

◉ PROFILE

Senior Software Engineer with 7+ years across industrial automation and railway systems, including 3+ years on railway Train Control & Monitoring Systems (TCMS) at Alstom. Experienced in C programming for TCMS middleware, resolving real-time issues on trains in operation through GDB core-dump analysis, LTTng trace analysis and root-cause diagnosis across software, hardware and network layers on Linux (Yocto, Debian). Works closely with the development team, including on-site collaboration in France.

◈ CORE EXPERTISE

C Middleware Development & Debugging
TCMS middleware layer · code flow · runtime behaviour

Core Dump & Trace Analysis
GDB core/crash dumps · LTTng traces · system logs

Train Communication Networks
TRDP · CIP · MVB · CAN · HPTS · Wireshark

Linux Platforms
Yocto · Debian · shell scripting · IPC

Railway Tooling
AGATE · ControlBuild · CODESYS · IEC 61131-3

Industrial Automation
Siemens PLC / SCADA · HMI · IoT integration

⚒ EXPERIENCE

Alstom Transport India Limited

JUL 2024 – PRESENT

Senior Software Engineer – TCMS (Redmine Product Team)

Bangalore

Part of the TCMS product team, resolving real-time software issues on metro trains in operation across international projects.

Middleware & Real-Time Debugging

- Develop and debug C code in the TCMS middleware layer – analysing code flow, runtime behaviour and fixing defects.
- Diagnose crashes and runtime faults using GDB core-dump analysis and LTTng trace analysis, isolating root causes across software, hardware and network layers.
- Debug real-time controller behaviour, IPC and protocol stacks (MVB, CIP, CAN, TRDP, HPTS) using Wireshark and system logs on Yocto/Debian-based platforms.
- Develop shell scripts to automate log collection from train controllers for faster fault analysis.

Development Team Collaboration

- Travelled twice to Alstom VBN, France (~1 month each) to work alongside the development team on issue analysis and fixes.
- Work closely with the development and V&V teams to review, validate and deliver fixes; use GitLab for code commits, merge requests and code reviews.
- Use IBM ClearCase / ClearQuest for version control, file management and change tracking.

Platforms & Tools

- Work on the AGATE platform (Alchemist, Diagnostics Editor, OCoCheTo, Dataplug Manager) and IEC 61131-3 / model-based tools (CODESYS, ControlBuild).

CONTROLLERS & HMI

- AGATE 4 – XPU
- AL3E-008
- Siemens SIMOTION D435 & D425
- S7-1500 · S7-1200 · S7-400H
- SIMATIC IoT2040 · Jetson Nano
- TP700 / TP900 Comfort, KTP400

EDUCATION

B.E. Electrical Engineering
Atmiya Institute of Technology & Science, Gujarat
2014 – 2018 · 83.5%

INTERNATIONAL WORK

Alstom VBN – France
2 visits, ~1 month each, working with the development team

Toulouse Metro – France
TCMS software

Bucharest Metro – Romania
TCMS software

ENEL, Bogotá – Colombia
Coal sampling gantry commissioning

LANGUAGES

English, Hindi, Gujarati, Sindhi

Grasim Industries Limited (Aditya Birla Group)
MAR 2023 – JUL 2024

Senior Engineer — Gujarat

- Commissioned an S7-400H redundant PLC system for the After Treatment Plant.
- Developed an application to acquire PLC data into MS SQL Server with automated daily report generation.
- Built PLC and SCADA systems to monitor and control chillers, compressors and field signals, consolidating data into a centralised platform with automated e-mail reporting.
- Converted manipulator program from ST to FBD in SIMATIC Manager for easier maintenance and fault-finding.

Alstom Transport India Limited
FEB 2022 – MAR 2023

Software Engineer — Bangalore

- Produced MPU ControlBuild software models compliant with TCMS Specification, Functional Design Document and Interface Control Document.
- Contributed to metro software for the Toulouse Metro (France) and Bucharest Metro (Romania) projects.
- Developed serial-communication software and validated it extensively using Docklight.
- Developed the ACC GUI application, using ControlBuild APIs to generate a complete project tree structure.
- Built and deployed the Patch Base Portal, a web application tracking packs and patches per project.
- Provided coding support for the Smart Metro platform.

Asmi Control System Pvt. Ltd.
FEB 2019 – FEB 2022

Project Engineer — Bangalore

SGS India – Hindalco Industries (Odisha & Madhya Pradesh)

- Commissioned automatic auger and secondary sampling systems: PLC programming, HMI and IoT program development applying Industry 4.0 concepts.

SGS Colombia – ENEL, Bogotá (Colombia)

- Commissioned a fully automatic coal sampling gantry system covering PLC programming, SCADA screens and server communication.

Grasim Industries – PSY & SSY Spinning Plants (Veraval, Gujarat)

- Retrofitted spinning machine drives with Siemens servo motors and PLC controllers; developed PLC logic and HMI screens.
- Developed SCADA software integrating 10+ spinning machines into a single monitoring system.

Walvoil Fluid Power · Comer Industries

- Automated component test-report generation on Node-RED with server storage, and on TP900 Comfort HMI using VBScript.

Simms Engineering Pvt. Ltd.
SEP 2018 – JAN 2019

Engineer (O&M) — Veraval, Gujarat

- Operated and maintained a 66 kV substation at Indian Rayon.

CAREER JOURNEY

2018	2019	2022	2023	2024 →
Engineer (O&M)	**Project Engineer**	**Software Engineer**	**Senior Engineer**	**Senior Software Engineer**
Simms Engineering	Asmi Control System	Alstom	Grasim Industries	Alstom

CAREER PATH

Electrical O&M → PLC / SCADA & IoT → Railway TCMS → **Embedded Linux Middleware & Real-Time Diagnostics**